$m\frac{l^2}{T^2}$. Past G, enter at O the field right of G.

Uniform magnitude B, into the page; all is in vacuum. Define the Cartesian axes shown with origin O: Oxy; boundary MN is parallel to the x axis. At $t = 0$, open D_1, D_2 together, admitting to C_1; at $t = 3T/4$, close D_1 to stop entry to C_1; at $t = 10T/4$ close D_2, stopping entry to G. The fastest particle from O passes point Q, at (3 cm,3 cm). **For the slowest particle entering at O also to pass Q, where in plane Oxy must MN be** shifted parallel to itself?

6. Shown: blocks a、b、c、d on a smooth level table,



all of mass m. a、b spring: natural length l, stiffness k_1; c、d are joined by a spring of natural length l, stiffness k_2. Their centres are collinear. b、c collisions are perfectly elastic and brief. Both springs start at natural length. c、d rest; a、b move right at velocity v_0. Discuss their motion quantitatively.

1. $k_1 = k_2$: how do all four move relative to the table?

2. $k_1 = 4k_2$: how do all four move relative to the table?

6. (25 points)

Blocks a、b、c、d, of mass m, lie on a smooth level table. A spring of natural length l joins a、b; its stiffness is k_1. A spring of natural length l, stiffness k_2, joins c、d. All four centres lie on a line. b、c collisions are perfectly elastic and very brief. Both springs initially have natural length; c、d rest, while a、b move right at the same velocity v_0. Discuss quantitatively:

（1） For $k_1=k_2$, how do the four blocks move relative to the table?

（2） For $k_1=4k_2$, how do the four blocks move relative to the table?